DAY 1 · THURSDAY 22 OCTOBER

# Welcome to ELELEM Chile

From LLMs to Agentic AI — three days to understand, not just do

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Harvard University

THE IDEA

# Understand, not just do

- Anyone can wire up an agent in ten lines of code. The hard part is knowing **why it works — and why it breaks**.
- So we go under the hood: how a language model predicts, what retrieval really adds, what turns an LLM into an agent, and how these systems fail.
- Every lecture is followed by a **lab** where you see the same idea in code.
- Ask questions at any time — interrupting is welcome.

**You will leave able to**

- Explain what an LLM does — and what it cannot do on its own
- Build a RAG pipeline and improve its retrieval
- Build tool-using agents, and know when one agent is enough
- Spot the risks, and design an evaluation for your own system

THREE DAYS

# From a language model to agentic systems

### Thursday · LLMs → RAG

- Language models: from n-grams to ChatGPT
- ChatGPT lab
- Retrieval-augmented generation
- Industry talk · RAG lab

### Friday · Agents

- Capstone brief — form teams
- Agents I: the loop, tools, MCP · lab
- Agents II: context, harnesses, skills · lab
- Industry talk · capstone worked example
- Pisco sour, from 18:00 🍸

### Saturday · Landscape

- The AI landscape & multi-agent systems
- Limitations, risks, evaluation
- Capstone reviews and discussion

Capstone: in teams, you design an agentic system — architecture and technical design, no code. Brief on Friday 9:00, designs due Saturday 9:00, reviews Saturday 11:20.

TODAY · THURSDAY 22 OCTOBER

# The plan for today

| Time | Session |
|---|---|
| **9:00–10:30** | Language models I — what a language model is, neural LMs, attention |
| **10:30–10:50** | Coffee |
| **10:50–12:30** | Language models II — transformers, BERT and GPT, from GPT to ChatGPT |
| **12:30–13:30** | Lunch |
| **13:30–15:10** | Lab · ChatGPT — from a base model to an assistant |
| **15:10–15:30** | Coffee |
| **15:30–17:00** | Retrieval-augmented generation |
| **17:00–17:30** | Industry talk · Ignacio Becker, CMPC |
| **17:30–18:00** | Lab · RAG |

THE TEAM

# Who is who

### Pavlos Protopapas

**Instructor**

Harvard University

### Guillermo Cabrera-Vives

**Instructor**

Universidad de Concepción

### Mabel Vidal

**Teaching team**

Universidad de Concepción

### Matías Olivas

**Teaching assistant**

Labs

### Karla Cerda

**Organizer**

Executive Director, CDIA

### Valentina Riveros

**Support team**

Logistics

Guest speakers: Ignacio Becker (CMPC) on Thursday · a Lead ML Engineer from CDIA on Friday

PRACTICALITIES

# Good to know

| | |
|---|---|
| **Wi-Fi** | UdeC Santiago · password udecsantiago |
| **Materials** | pavlosprotopapas.github.io/elelem-chile-2026 — slides and Colab notebooks, posted before each session |
| **Breaks** | Coffee 10:30 and 15:10 · lunch 12:30–13:30 |
| **Logistics** | Karla Cerda and Valentina Riveros (CDIA) · cdia@inf.udec.cl |
| **Language** | Lectures and materials in English |

Scan for the workshop page

BEFORE THE FIRST LAB (13:30)

# Run the readiness check

1

### Open the check

Workshop page → Before you arrive → Open in Colab

2

### Save a copy

File → Save a copy in Drive

3

### Add your key

🔑 Secrets → OPENAI_API_KEY (about USD 10 of credit)

4

### Run all cells

The last line prints ✅ ready

**No ✅ ready? Bring the output to any of us at the coffee break — before the first lab.**